

FILTER DESIGN TEST CASE

Notch Filter - Band 5 (850 MHz) for Public Safety Band Filter
Design Validation and Performance Characterization

DOCUMENT INFORMATION

Document Number:	FD-TC-0018
Revision:	Rev A.8
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Created Date:	2024-10-03
Last Modified:	2024-11-13
Status:	FAIL
Priority:	P1 - Critical

1. OBJECTIVE

This document describes the design, simulation, and characterization of a Notch Filter filter targeting Band 5 (850 MHz) for Public Safety Band Filter. The filter is designed on 128° YX LiNbO3 substrate with a target center frequency of 1025.7 MHz and bandwidth of 312.3 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the Public Safety Band Filter module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Notch Filter
- Frequency Band: Band 5 (850 MHz)
- Application: Public Safety Band Filter
- Substrate: 128° YX LiNbO3
- Package: Flip-Chip BGA
- Design Tool: Sonnet EM Suite v16
- Design Center: Cedar Rapids Lab

This specification applies to all variants of the Notch Filter design targeting Band 5 (850 MHz). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	1025.7 MHz
Bandwidth (3 dB):	312.3 MHz
Insertion Loss Target:	<= 2.1 dB
Return Loss Target:	>= 12.0 dB
Out-of-Band Rejection:	>= 39.0 dB
Isolation (Duplexer):	>= 41.0 dB
Q Factor:	9246
Temperature Coefficient:	-17.9 ppm/°C
Die Size:	2.0 x 2.0 mm
Wafer Size:	8-inch
Number of Resonators:	9
Electrode Material:	Pt/Al

Passivation: Si3N4

4. TEST PROCEDURE

1. Open Sonnet EM Suite v16 and load the Notch Filter template project.
2. Define the substrate stack: 128° YX LiNbO3 with specified layer thicknesses.
3. Set design targets: center freq 1025.7 MHz, BW 312.3 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.1 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (Flip-Chip BGA).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, 128° YX LiNbO3).
10. Perform on-wafer probing using Anritsu MS46122B ShockLine.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 3077 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.1 dB
- Return loss in passband shall be >= 12.0 dB
- Out-of-band rejection at +/- 468 MHz offset >= 39.0 dB
- Group delay variation across passband shall be <= 10 ns
- Temperature drift over -40°C to +85°C shall be <= 2.3 MHz
- Power handling shall be >= +31 dBm at 1 dB compression
- ESD tolerance >= 2000 V (HBM)
- Die yield on qualification lot >= 91%

6. TEST RESULTS

Measurement Date: 2024-11-13
Devices Tested: 72
Insertion Loss (meas): 1.79 dB
Return Loss (meas): 13.6 dB
Rejection (meas): 41.8 dB
Isolation (meas): 45.8 dB
Q Factor (meas): 9246
Group Delay Variation: 2.2 ns
Power Handling: +26 dBm
Die Yield: 95.7%

Result Classification: FAIL

7. OBSERVATIONS AND FINDINGS

- a) The Notch Filter design demonstrated below-target performance for Band 5 (850 MHz).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Wafer-level uniformity was excellent (sigma < 1.5%).

8. CORRECTIVE ACTIONS

- 1. Optimize resonator geometry to reduce insertion loss by ~0.3 dB.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Escalate to advanced materials team for alternative piezoelectric stack.

9. SIGN-OFF

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